

Supplementary Material

A Comparative Analysis of Support Vector Machines and Multilayer Perceptrons on the Spambase Dataset

A. Glossary of Key Terms

- **Cross-Validation (CV):** A method used to evaluate model performance by splitting training data into multiple folds and averaging results.
- **Validation Set:** A subset of data used after cross-validation to further tune model configurations.
- **Test Set:** A separate dataset used only once for final evaluation on unseen data.
- **Overfitting:** When a model performs well on validation data but does not generalise to test data.
- **Feature Scaling:** Standardising input features to ensure consistent ranges across variables.
- **Optimizer:** Algorithm used to update model parameters during training (e.g., Adam, SGD with momentum).
- **False Negatives (FN):** Positive instances incorrectly predicted as negative.
- **False Positives (FP):** Negative instances incorrectly predicted as positive.

B. Intermediate Results and Model Selection

(i) SVM Hyperparameter Selection (Cross-Validation):

Different values of C and gamma were evaluated.

- C = 0.1 showed lower performance
- C = 1 and C = 10 performed better
- C = 10 with gamma = scale achieved the best results

Final SVM configuration:

C = 10, gamma = scale

```
==== SVM SUMMARY ====
C=0.1, gamma=scale - Mean: 0.9073, Std: 0.0120
C=1, gamma=scale - Mean: 0.9318, Std: 0.0082
C=10, gamma=scale - Mean: 0.9348, Std: 0.0056
C=1, gamma=0.01 - Mean: 0.9291, Std: 0.0095
C=1, gamma=0.1 - Mean: 0.9111, Std: 0.0138
```

Fig 1: Cross Validation of SVM

(ii) MLP Architecture Selection (Cross-Validation):

Two architectures were compared:

- 2 hidden layers
- 3 hidden layers

The 3-layer model achieved higher mean validation accuracy but showed slightly higher variation. It was selected for further optimisation.

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==== SUMMARY ====
2_layer_lr_0.05 - Mean: 0.9402, Std: 0.0049
2_layer_lr_0.01 - Mean: 0.9304, Std: 0.0051
3_layer_lr_0.05 - Mean: 0.9416, Std: 0.0062
3_layer_lr_0.01 - Mean: 0.9375, Std: 0.0058
```

Fig 2: Cross Validation of MLP

(iii) MLP Optimisation (Validation Split):

Different optimizers were evaluated:

- Adam (baseline)
- Adam + weight decay
- SGD + momentum
- SGD + momentum + weight decay

===== FINAL RESULTS (Repeated Runs) =====				
	Model	Mean Validation Accuracy	Std Dev	
0	Adam (baseline)	0.990761	0.001584	
1	Adam + weight decay	0.986685	0.003686	
2	SGD + momentum	0.993750	0.000666	
3	SGD + momentum + weight decay	0.990489	0.001215	

Fig 4: MLP Optimization Result

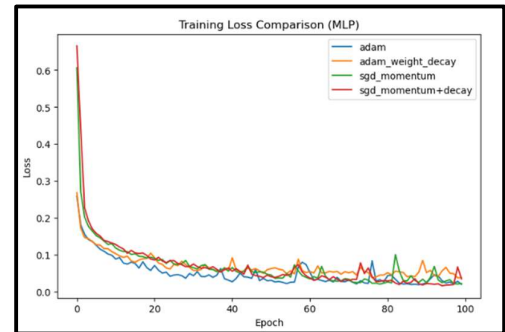


Fig 5: Training Loss Chart

The training loss curves in Fig 5 show that all optimizers converge steadily, with Adam achieving faster initial convergence, while SGD with momentum provides more stable behaviour in later epochs. Despite similar final loss values, SGD with momentum demonstrates smoother convergence, supporting its selection as the final model. Results showed that Adam performed best during cross-validation (Fig 2) but SGD with momentum achieved higher validation accuracy during further tuning (Fig 4).

Final MLP configuration: SGD with momentum (learning rate = 0.01)

C. Negative and Intermediate Findings

- The optimizer selected through cross-validation (Adam) did not yield the best final performance. SGD with momentum performed better during further validation, indicating that cross-validation alone was not sufficient for final model selection.
- The 3-layer MLP improved performance compared to the 2-layer model but showed slightly higher variability, indicating a trade-off between performance and stability.
- The validation accuracy of the MLP (~0.9938) was significantly higher than the test accuracy (~0.9327), indicating that validation performance can be optimistic after multiple rounds of tuning.

D. Implementation Issues and Fixes

- **Scaling inconsistency:** Incorrect or inconsistent scaling of test data led to degraded performance. This was resolved by ensuring the same scaler was applied consistently across all datasets.
- **Model selection consistency:** Differences between cross-validation and validation results required careful tracking to ensure the correct final model configuration was used.
- **Tensor shape mismatch:** Errors during evaluation were caused by mismatched tensor shapes. This was fixed by reshaping predictions and labels consistently.

AI Assistance Declaration: I declare that the project objective, experimental design, and data analysis presented in this work are my own. I utilized AI tools to assist with structuring parts of the code, debugging implementation issues, and refining the wording of the report and supplementary material. All key decisions, including model selection, architecture design, hyperparameter tuning, preprocessing, and evaluation of results, were carried out independently. The interpretation of results and conclusions presented in this project reflect my own understanding of the methods and findings.